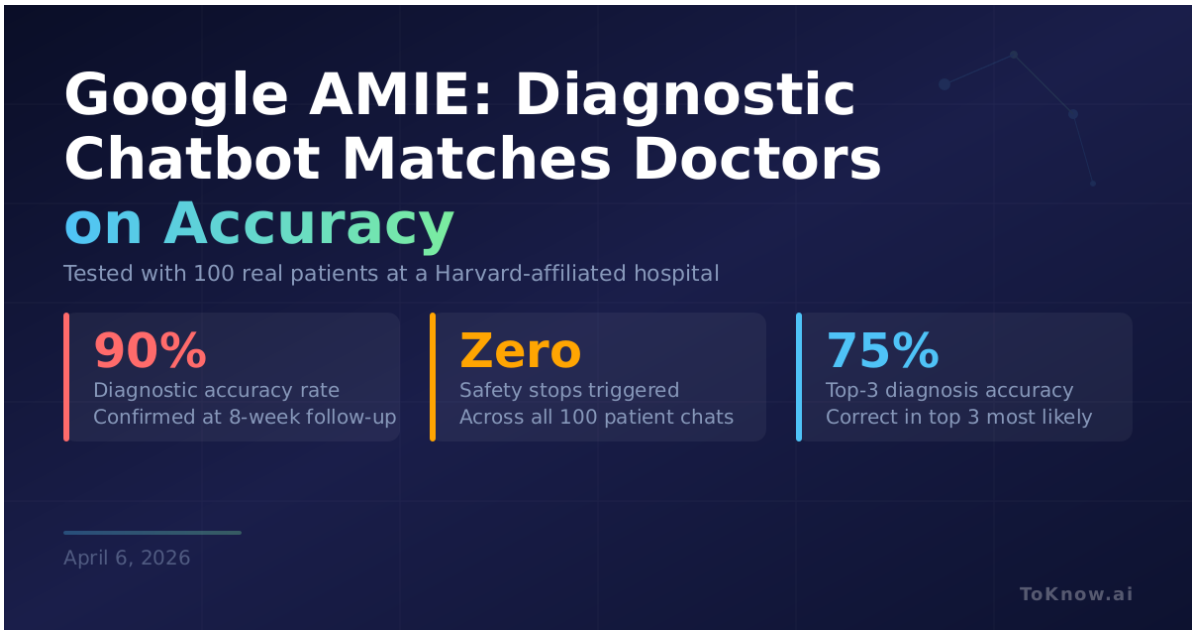
Google AMIE: Diagnostic Chatbot Tested with 100 Real Patients Matches Doctors on Accuracy

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Google tested its conversational diagnostic AI, AMIE, with 100 real patients at Beth Israel Deaconess Medical Center (Harvard-affiliated) in a prospective clinical feasibility study. Previous conversational AI studies used simulated patients or were retrospective; this is the first

prospective study of an LLM-based diagnostic chatbot in a live primary care clinic with real-time physician oversight. Patients chatted with AMIE via text before their primary care appointment while a physician monitored live, ready to intervene on four pre-defined safety criteria. None were triggered. AMIE generated a transcript and summary for the doctor to review before the visit. A blinded panel of evaluators rated AMIE's differential diagnoses (ranked lists of possible conditions) and management plans on par with the primary care physicians (PCPs). AMIE matched the correct final diagnosis, confirmed via chart review eight weeks later, in 90% of cases, with 75% top-3 accuracy. PCPs only outperformed AMIE on practicality and cost of management plans, expected since AMIE lacked access to medical records or physical exams.

Doctors spend a large chunk of patient visits just gathering basic history and symptoms. PCPs who reviewed AMIE's pre-visit summaries reported their consultations shifted from data collection to data verification and shared decision-making, which is the high-value part of a visit. Patient attitudes toward AI improved significantly after interacting with AMIE ($p < 0.001$), countering fears of technology rejection. For a healthcare system stretched thin on primary care providers, an AI that reliably takes patient history and gets the diagnosis right 90% of the time could meaningfully free up clinician time.

This is a single-center feasibility study, not a controlled efficacy trial, and Google has committed to larger controlled follow-ups. But the direction is clear: clinical AI is moving from simulated benchmarks to real patients. The bottleneck is rigorous evidence generation, not technical capability.

Sources:

- Google Research Blog: AMIE Clinical Study
- Paper: Prospective Clinical Feasibility Study (arXiv)
- Clinical Trial Registration (NCT06911398)
- AMIE Clinician Study (Nature)
- AMIE Patient Actor Study (Nature)

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